MOTIFDIFF-EM: INDUCTIVE SEMI-SUPERVISED MOLECULAR PROPERTY PREDICTION VIA ANALYTIC MOTIF DIFFUSION DISTILLATION

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ABSTRACT

Molecular property prediction is typically formulated as learning a graph-to-label function on a set of molecules treated as independent samples. However, molecules in a dataset often share chemically meaningful substructures (e.g., rings and functional groups), and these shared motifs induce a latent *inter-molecule* dependency that standard molecular GNN pipelines under-exploit. Meanwhile, directly learning on a dataset-level graph of molecules is usually *transductive* (requiring the test molecules to be connected at inference), which is misaligned with common *inductive* evaluation protocols in molecular benchmarks. In this work, we propose an **inductive semi-supervised** framework that explicitly leverages dataset-level motif sharing during training while preserving graph-independent inference at test time. We first construct a motif-induced molecular graph where nodes are molecules and edges encode motif overlap with reliability-aware weighting. On this graph, we design an *analytic motif-diffusion teacher* that produces calibrated soft pseudo-labels by minimizing a convex objective combining supervised likelihood and smoothness regularization. We then distill these pseudo-labels into an arbitrary local molecular encoder (e.g., GIN) via a variational EM procedure that alternates between (E) updating teacher pseudo-labels and (M) optimizing the student on labeled and pseudo-labeled data. The resulting method (i) exploits cross-molecule structure sharing, (ii) remains inductive at inference, and (iii) yields motif-level attributions that improve interpretability.

1 INTRODUCTION

Predicting molecular properties from graph-structured representations is a core task in drug discovery and materials design. Graph neural networks (GNNs) and message passing architectures have become a dominant approach due to their ability to learn task-adaptive representations from atom–bond graphs Gilmer et al. (2017); Xu et al. (2019). Large benchmark suites such as MoleculeNet have standardized evaluation across diverse molecular endpoints, but also highlight persistent challenges under label scarcity, distribution shift, and structure–property discontinuities Wu et al. (2018); van Tilborg et al. (2022).

A key yet under-utilized statistical property of molecular datasets is that molecules are *not* i.i.d. in the sense of structural primitives: many compounds share recurring motifs (rings, functional groups, scaffolds), and these motifs are often directly tied to properties. Treating each molecule in isolation discards potentially useful information: if two molecules share multiple reliable motifs, their labels are likely correlated; conversely, rare or noisy motifs should contribute less. This motivates augmenting molecular learning with *dataset-level relational structure* induced by motif sharing.

Several recent works attempt to exploit inter-molecule relations by constructing heterogeneous motif graphs or similarity graphs and performing message passing across molecules Yu & Gao (2022); Yao et al. (2024); Zhao et al. (2024). While effective, these approaches typically inherit a *transductive* dependency: predictions for test molecules may rely on their placement in the global graph, which is unavailable or ill-defined when new molecules arrive after training. This mismatch is especially problematic in molecular property prediction, where realistic deployment is inherently inductive.

We address this gap with a framework that uses dataset-level structure *only during training* and distills it into a graph-independent predictor for inference. Concretely, we build a motif-induced molecular graph over training molecules, then solve an *analytic diffusion* problem on this graph to obtain soft pseudo-labels. The diffusion teacher is obtained by minimizing a convex objective that combines (i) supervised likelihood on labeled molecules and (ii) a smoothness regularizer that propagates information along reliable motif-sharing edges. We then distill the teacher into a local molecular encoder (any message-passing GNN, with GIN as a default instantiation) using a *variational EM* procedure that alternates between updating pseudo-labels (E-step) and training the student (M-step), similar in spirit to EM-style co-training frameworks Qu et al. (2019); Zhao et al. (2023) but specialized to molecular graphs and motif-based relational inductive bias.

Contributions. (i) We formulate **motif-induced inductive semi-supervision** for molecular property prediction: leveraging cross-molecule motif sharing during training without requiring a test-time dataset graph. (ii) We propose an **analytic motif-diffusion teacher** with reliability-aware edge weighting and a convex training objective, producing calibrated soft pseudo-labels. (iii) We develop a **variational EM distillation** algorithm that transfers dataset-level relational information into an arbitrary local molecular encoder, enabling standard inductive evaluation. (iv) The teacher yields **motif-level attributions** (via edge/motif weights and diffusion influence), improving interpretability and supporting diagnosis under challenging regimes such as activity cliffs van Tilborg et al. (2022).

2 Related Work

Molecular property prediction with GNNs. Message passing neural networks and their variants learn molecular representations by iteratively aggregating neighborhood information on atom–bond graphs Gilmer et al. (2017). Architectures such as GIN improve expressive power and have become common baselines in molecular benchmarks Xu et al. (2019). Benchmark suites (e.g., MoleculeNet) consolidate datasets and protocols, but also reveal limitations under small data and difficult structure–activity landscapes Wu et al. (2018); van Tilborg et al. (2022).

Exploiting inter-molecule relations. A line of work constructs dataset-level graphs to share information across molecules. HM-GNN builds a heterogeneous graph of molecule nodes and motif nodes with learned message passing Yu & Gao (2022). MSSM-GNN uses molecule–molecule similarity graphs (e.g., via kernels) to incorporate global structural similarity Yao et al. (2024). GSL-MPP explores structure learning to better capture relationships for molecular property prediction Zhao et al. (2024). These methods demonstrate that cross-molecule structure is useful, but they often couple prediction to the availability of the global graph at inference.

Graph diffusion and semi-supervised propagation. Graph diffusion/propagation methods (e.g., PPNP/APPNP) use personalized PageRank-style operators to exploit long-range neighborhoods and improve stability Klicpera et al. (2019). Correct & Smooth further shows that simple base predictors can be substantially improved by label-propagation-like post-processing in transductive node classification Huang et al. (2021). Our teacher is closely related in spirit—it uses diffusion over a dataset-level graph—but is tailored to motif-induced molecule graphs and is explicitly used as a *training-only* teacher for inductive distillation. In the molecular domain, Geo-DEG performs diffusion over a learned grammar-induced geometry for data-efficient property prediction Guo et al. (2023); unlike such representation/geometry diffusion, our diffusion operates on a dataset-level motif-sharing graph and is used only during training for inductive distillation.

Inductive distillation and EM-style co-training. Graph-independent inference via distilling a structure-aware teacher into a structure-free student has been explored primarily in node-level settings, e.g., GKD Ghorbani et al. (2021) and Propagate & Distill Shin & Shin (2023). Our setting differs in four aspects: (i) labels are at the molecule (graph) level, (ii) the relational graph is *constructed* from interpretable motif sharing (with reliability weighting) rather than given as an input graph, (iii) the teacher is a parameter-free convex diffusion solver with explicit motif-level attributions, and (iv) inference uses only a local molecular encoder without any dataset graph. Separately, GLEM formalizes alternating optimization of graph and text modules via variational EM on large text-attributed graphs Zhao et al. (2023). Our method aligns with this general philosophy but differs in the domain (molecular graphs), the relational construction (motif-induced molecule graph with reliability weighting), and the analytic convex teacher design. EM-style semi-supervised co-training further alternates between refining latent labels and updating a predictor, as in GMNN Qu et al. (2019) and GLEM Zhao et al. (2023); in contrast, our E-step admits an analytic fixed point derived from convex label inference on the motif-induced graph, while the M-step trains a backbone-agnostic molecular GNN by distilling these targets.

3 METHOD

3.1 PROBLEM SETUP

We study semi-supervised molecular property prediction. Let $\mathcal{D}_L = \{(G_i, y_i)\}_{i=1}^{n_L}$ be labeled molecules and $\mathcal{D}_U = \{G_i\}_{i=n_L+1}^{n_L+n_U}$ unlabeled molecules, with $N = n_L + n_U$. Each molecule is a graph $G_i = (V_i, E_i, \mathbf{X}_i^v, \mathbf{X}_i^e)$.

Main-text presentation (multi-class). In the main text we present a C-class classification task ($y_i \in \{1, \dots, C\}$). Binary classification is the special case $C = 2$. Regression instantiation is provided in the Appendix.

Multi-task and missing labels. For T tasks, task t has C_t classes and labels y_{it} may be missing. We use a mask $m_{it} \in \{0, 1\}$ indicating whether y_{it} is observed. For each task t, define the observed labeled index set

$$\mathcal{L}_t = \{i \in \{1, \dots, N\} : i \leq n_L \ \wedge \ m_{it} = 1\}, \qquad \mathcal{U}_t = \{1, \dots, N\} \setminus \mathcal{L}_t. \tag{1}$$

Thus a labeled molecule with missing task-t label is treated as unlabeled for task t.

Goal (inductive). At test time, predictions for a molecule G depend only on G and learned parameters, without access to other test molecules and without diffusion on a test-time inter-molecule graph.

3.2 OVERVIEW

MotifDiff-EM consists of: (i) a **local** molecular predictor (student) f_θ operating on a single molecule graph, and (ii) a **parameter-free motif-diffusion teacher** that performs label inference on a graph of molecules connected by shared, interpretable motifs. Training alternates between:

- **Teacher step:** infer soft targets $\{\mathbf{Q}^{(t)}\}$ by solving a diffusion fixed point.
- **Student step:** update θ by minimizing supervised loss on observed labels plus a distillation loss matching $\{\mathbf{Q}^{(t)}\}$ on unlabeled molecules (with confidence gating).

At inference, only the student f_θ is used.

3.3 LOCAL MOLECULAR PREDICTOR (BACKBONE-AGNOSTIC)

The student encodes a molecule graph and predicts a categorical distribution:

$$\mathbf{H}_i = \mathrm{Enc}_\theta(G_i) \in \mathbb{R}^{|V_i| \times d}, \tag{2}$$

$$\mathbf{z}_i = \mathrm{Pool}(\mathbf{H}_i) \in \mathbb{R}^d, \tag{3}$$

$$\mathbf{p}_{it} = p_\theta(\cdot \mid G_i, t) = \mathrm{Softmax}(\mathbf{W}_t \mathbf{z}_i) \in \Delta^{C_t - 1}. \tag{4}$$

We instantiate Enc_θ with GIN, but the framework only requires $\mathbf{p}_{it}$, so the encoder is replaceable.

Cross-entropy losses. Hard-label cross entropy:

$$\mathrm{CE}(y, \mathbf{p}) = -\log p_y. \tag{5}$$

Soft-label cross entropy (used for distillation):

$$\mathrm{CE}(\mathbf{q}, \mathbf{p}) = -\sum_{c=1}^{C} q_c \log p_c. \tag{6}$$

Supervised objective (multi-task, masked).

$$\mathcal{L}_{\text{sup}}(\theta) = \sum_{t=1}^{T} \sum_{i \in \mathcal{L}_t} \text{CE}(y_{it}, \mathbf{p}_{it}) \,. \tag{7}$$

3.4 Interpretable Motifs and Molecule–Motif Incidence

Motifs provide interpretable shared substructures and define the global relational structure among molecules. For each molecule G_i, we extract a motif set $\mathcal{M}(i)$ using: (i) Bemis–Murcko scaffold, (ii) functional-group SMARTS templates, (iii) optional BRICS fragments.

Frequency filtering. Let $\text{df}(m) = |\{i : m \in \mathcal{M}(i)\}|$. We prune motifs by

$$\tau_{\min} \leq \text{df}(m) \leq \tau_{\max} N, \tag{8}$$

to remove extremely rare motifs and ubiquitous motifs that induce degree bias.

IDF weighting. For remaining motifs indexed by $m \in \{1, \ldots, M\}$, define

$$\text{IDF}(m) = \log \frac{N}{\text{df}(m) + \epsilon}. \tag{9}$$

3.4.1 Cliff-Aware Motif Reliability Reweighting

Diffusion over an inter-molecule graph implicitly assumes a degree of label homophily. In molecular property prediction, this assumption can be violated due to *activity cliffs* (similar substructures but different properties). We therefore downweight motifs whose labeled occurrences exhibit high label inconsistency.

Motif reliability from labeled label-entropy (single-task multi-class). Consider the main-text C-class setting. For a motif m, let $\mathcal{S}(m) = \{i \in \mathcal{D}_L : m \in \mathcal{M}(i)\}$ and $n_m = |\mathcal{S}(m)|$. Define a smoothed empirical class distribution

$$\pi_{mc} = \frac{\sum_{i \in \mathcal{S}(m)} \mathbb{I}[y_i = c] + \beta}{n_m + C\beta}, \qquad c = 1, \ldots, C, \tag{10}$$

where $\beta > 0$ is a Laplace smoothing constant. Define motif label entropy

$$H_m = -\sum_{c=1}^{C} \pi_{mc} \log(\pi_{mc} + \epsilon), \tag{11}$$

and motif reliability

$$r_m = \exp(-\gamma H_m) \in (0, 1], \qquad \gamma \geq 0. \tag{12}$$

The shared reliability of multi-task with missing labels is provided in the Appendix.

Reliability-weighted incidence matrix. We incorporate reliability by defining the incidence matrix $\mathbf{B} \in \mathbb{R}^{N \times M}$ as

$$\mathbf{B}_{im} = \mathbb{I}[m \in \mathcal{M}(i)] \cdot \text{IDF}(m) \cdot r_m. \tag{13}$$

All subsequent teacher definitions are computed from this reweighted $\mathbf{B}$.

3.5 Motif-Diffusion Teacher

The teacher performs convex label inference over a motif-induced molecule graph. Crucially, the teacher has *no trainable parameters*.

3.5.1 Motif-induced symmetric molecule graph

Define motif degrees and a symmetric molecule–molecule weight matrix:

$$\mathbf{D}_M = \text{diag}(\mathbf{B}^\top \mathbf{1}) \in \mathbb{R}^{M \times M}, \qquad \mathbf{W} = \mathbf{B}\mathbf{D}_M^{-1}\mathbf{B}^\top \in \mathbb{R}^{N \times N}. \tag{14}$$

Intuitively, W_{ij} aggregates shared motifs between molecules i and j, normalized by motif degrees.

Define molecule degrees and the (combinatorial) Laplacian:

$$\mathbf{D} = \text{diag}(\mathbf{W}\mathbf{1}) \in \mathbb{R}^{N \times N}, \qquad \mathbf{L} = \mathbf{D} - \mathbf{W}. \tag{15}$$

Row-stochastic random walk. Define the random-walk transition

$$\mathbf{P} = \mathbf{D}^{-1}\mathbf{W}. \tag{16}$$

To ensure $\mathbf{P}$ is row-stochastic on *all* nodes, we adopt the convention: if $D_{ii} = 0$ (isolated after filtering), set $P_{ii} = 1$ and $P_{ij} = 0$ for $j \neq i$. This yields $\sum_j P_{ij} = 1$ for every row and guarantees well-defined diffusion for all molecules.

3.5.2 Anchors (multi-class, task-wise)

For each task t with C_t classes, define anchor distributions $\tilde{\mathbf{Y}}^{(t)} \in \mathbb{R}^{N \times C_t}$:

$$\tilde{\mathbf{Y}}_i^{(t)} = \begin{cases} \text{onehot}(y_{it}), & i \in \mathcal{L}_t, \\ \mathbf{p}_{it}, & i \in \mathcal{U}_t. \end{cases} \tag{17}$$

Anchors inject ground-truth on observed labeled entries and use the student's current belief otherwise.

3.5.3 Diffusion fixed point (teacher inference)

For each task t, the teacher target $\mathbf{Q}^{(t)} \in \mathbb{R}^{N \times C_t}$ is defined as the unique solution of

$$\mathbf{Q}^{(t)} = (1-\alpha)\tilde{\mathbf{Y}}^{(t)} + \alpha \mathbf{P}\mathbf{Q}^{(t)}, \qquad \alpha \in (0,1). \tag{18}$$

We compute it by power iteration:

$$\mathbf{Q}_{k+1}^{(t)} = (1-\alpha)\tilde{\mathbf{Y}}^{(t)} + \alpha \mathbf{P}\mathbf{Q}_k^{(t)}, \qquad \mathbf{Q}_0^{(t)} = \tilde{\mathbf{Y}}^{(t)}. \tag{19}$$

3.5.4 Confidence gating

For multi-class tasks, we distill only confident unlabeled targets:

$$g_{it} = \mathbb{I}\Big[\max_c Q_{ic}^{(t)} \geq \tau\Big], \qquad \tau \in (0,1). \tag{20}$$

3.6 Student Update (Explicit Distillation Objective)

Given $\{\mathbf{Q}^{(t)}\}_{t=1}^T$, the student minimizes

$$\min_\theta \ \mathcal{L}_{\text{sup}}(\theta) + \lambda \sum_{t=1}^T \sum_{i \in \mathcal{U}_t} g_{it} \cdot \text{CE}\Big(\mathbf{Q}_i^{(t)}, \mathbf{p}_{it}\Big), \tag{21}$$

where $\lambda > 0$ balances supervised and distillation terms.

Inductive inference. At test time, we discard the teacher and output $\mathbf{p}_{\text{test},t} = p_\theta(\cdot \mid G_{\text{test}}, t)$.

3.7 Motif-Level Attribution for Teacher Predictions

A key benefit of motif-induced diffusion is that teacher predictions admit an explicit decomposition into contributions from (i) anchor terms, (ii) neighbor molecules, and (iii) individual motifs.

Neighbor-molecule contribution (one-step). Fix a task t and drop (t) for readability. From the diffusion fixed point $\mathbf{Q} = (1-\alpha)\tilde{\mathbf{Y}} + \alpha\mathbf{P}\mathbf{Q}$, for any molecule i and class c:

$$Q_{ic} = (1-\alpha)\tilde{Y}_{ic} + \sum_{j=1}^N \underbrace{\alpha P_{ij} Q_{jc}}_{\text{contribution from molecule } j}. \tag{22}$$

Thus the scalar $\alpha P_{ij} Q_{jc}$ quantifies how much neighbor molecule j supports class c at molecule i.

Motif contribution (one-step, exact decomposition). Recall $\mathbf{P} = \mathbf{D}^{-1}\mathbf{B}\mathbf{D}_M^{-1}\mathbf{B}^\top$. Define intermediate motif-aggregated score

$$U_{mc} := \sum_{j=1}^{N} \frac{B_{jm}}{(D_M)_{mm}} Q_{jc}, \qquad m = 1, \dots, M. \tag{23}$$

Then

$$(\mathbf{PQ})_{ic} = \sum_{m=1}^{M} \frac{B_{im}}{D_{ii}} U_{mc}. \tag{24}$$

Define the motif-level contribution score

$$s_{imc} := \alpha \cdot \frac{B_{im}}{D_{ii}} U_{mc}, \qquad \text{so that} \quad Q_{ic} = (1-\alpha)\tilde{Y}_{ic} + \sum_{m=1}^{M} s_{imc}. \tag{25}$$

Therefore, the top-k motifs by s_{imc} provide an explicit explanation for the teacher's belief on class c.

Multi-hop attribution (optional). Using Neumann expansion, $\mathbf{Q} = (1-\alpha)\sum_{k=0}^{\infty}(\alpha\mathbf{P})^k\tilde{\mathbf{Y}}$, we can attribute predictions to length-k walks on the motif-induced graph. in practice, truncating to small K yields stable explanations.

3.8 Training Algorithm

Algorithm 1 MotifDiff-EM training (teacher diffusion + student distillation).

1: **Input:** $\mathcal{D}_L, \mathcal{D}_U$; masks m_{it}; α, τ, λ; rounds R; diffusion iters K.
2: Extract motifs; compute $\mathrm{df}(\cdot)$ and $\mathrm{IDF}(\cdot)$ (Eq. 9); compute r_m (Eq. 34).
3: Build $\mathbf{B}$ by Eq. 13; compute $\mathbf{D}_M, \mathbf{W}, \mathbf{D}, \mathbf{P}$ (Eq. 14–16).
4: Initialize θ by minimizing $\mathcal{L}_{\text{sup}}(\theta)$ (Eq. 7) on $\mathcal{D}_L$.
5: **for** $r = 1$ to R **do**
6:     **Teacher:** build anchors $\tilde{\mathbf{Y}}^{(t)}$ for all t (Eq. 17).
7:     **Teacher:** for each task t, run K iterations of Eq. 19 to obtain $\mathbf{Q}^{(t)}$.
8:     **Gating:** compute g_{it} on $\mathcal{U}_t$ via Eq. 20.
9:     **Student:** update θ by minimizing Eq. 21 with minibatches.
10: **end for**
11: **Return:** student model f_θ (used at inference).

3.9 Complexity

Let $\mathrm{nnz}(\mathbf{B})$ be the number of nonzeros. Each multiplication by $\mathbf{P}$ can be implemented as sparse bipartite propagation (molecule→motif→molecule) and costs $O(\mathrm{nnz}(\mathbf{B}))$ per iteration, per task. The teacher is thus lightweight compared to training a global GNN on an inter-molecule graph. The student step is standard molecular GNN training.

4 Analysis

4.1 Basic Properties of the Motif-Induced Graph

Lemma 1 (Symmetry and PSD of $\mathbf{W}$). *Let $\mathbf{W} = \mathbf{B}\mathbf{D}_M^{-1}\mathbf{B}^\top$ with $\mathbf{D}_M \succ 0$. Then $\mathbf{W}$ is symmetric and positive semidefinite (PSD).*

Proof. Symmetry: $\mathbf{W}^\top = (\mathbf{B}\mathbf{D}_M^{-1}\mathbf{B}^\top)^\top = \mathbf{B}\mathbf{D}_M^{-1}\mathbf{B}^\top = \mathbf{W}$. PSD: for any $\mathbf{x} \in \mathbb{R}^N$,

$$\mathbf{x}^\top\mathbf{W}\mathbf{x} = \mathbf{x}^\top\mathbf{B}\mathbf{D}_M^{-1}\mathbf{B}^\top\mathbf{x} = (\mathbf{B}^\top\mathbf{x})^\top\mathbf{D}_M^{-1}(\mathbf{B}^\top\mathbf{x}) \geq 0,$$

since $\mathbf{D}_M^{-1}$ is diagonal with nonnegative entries. $\square$

Lemma 2 (Reliability reweighting preserves symmetry and PSD)**.** *Let* $\mathbf{B}' = \mathbf{B}_0\mathbf{R}$ *where* $\mathbf{R} = \mathrm{diag}(r_1, \ldots, r_M)$ *with* $r_m \geq 0$*, and define* $\mathbf{W}' = \mathbf{B}'(\mathbf{D}'_M)^{-1}(\mathbf{B}')^\top$ *where* $\mathbf{D}'_M = \mathrm{diag}((\mathbf{B}')^\top\mathbf{1})$*. Then* $\mathbf{W}'$ *is symmetric PSD. Consequently,* $\mathbf{L}' = \mathbf{D}' - \mathbf{W}'$ *is symmetric PSD, where* $\mathbf{D}' = \mathrm{diag}(\mathbf{W}'\mathbf{1})$*.*

Proof. Symmetry follows from transpose invariance: $\mathbf{W}'^\top = \mathbf{B}'(\mathbf{D}'_M)^{-1}(\mathbf{B}')^\top = \mathbf{W}'$. For PSD, for any $\mathbf{x} \in \mathbb{R}^N$,

$$\mathbf{x}^\top\mathbf{W}'\mathbf{x} = (\mathbf{B}'^\top\mathbf{x})^\top(\mathbf{D}'_M)^{-1}(\mathbf{B}'^\top\mathbf{x}) \geq 0,$$

since $(\mathbf{D}'_M)^{-1}$ is diagonal with nonnegative entries on the retained motif vocabulary. The Laplacian PSD then follows from the standard energy identity (Lemma 3 below). □

Lemma 3 (Laplacian energy identity and PSD)**.** *Let* $\mathbf{L} = \mathbf{D} - \mathbf{W}$ *with* $\mathbf{D} = \mathrm{diag}(\mathbf{W}\mathbf{1})$*. Then* $\mathbf{L}$ *is symmetric PSD and for any* $\mathbf{U} \in \mathbb{R}^{N\times C}$*,*

$$\mathrm{Tr}(\mathbf{U}^\top\mathbf{L}\mathbf{U}) = \frac{1}{2}\sum_{i,j} W_{ij}\|\mathbf{U}_i - \mathbf{U}_j\|_2^2 \;\geq\; 0. \tag{26}$$

Proof. Symmetry follows from symmetry of $\mathbf{W}$ and diagonality of $\mathbf{D}$. For the identity, expand:

$$\sum_{i,j} W_{ij}\|\mathbf{U}_i - \mathbf{U}_j\|_2^2 = \sum_{i,j} W_{ij}(\|\mathbf{U}_i\|_2^2 + \|\mathbf{U}_j\|_2^2 - 2\mathbf{U}_i^\top\mathbf{U}_j).$$

Because $W_{ij} = W_{ji}$,

$$\sum_{i,j} W_{ij}\|\mathbf{U}_i\|_2^2 = \sum_i \|\mathbf{U}_i\|_2^2 \sum_j W_{ij} = \sum_i D_{ii}\|\mathbf{U}_i\|_2^2 = \mathrm{Tr}(\mathbf{U}^\top\mathbf{D}\mathbf{U}),$$

and similarly for the $\|\mathbf{U}_j\|_2^2$ term, giving another identical contribution. The cross term is $2\sum_{i,j} W_{ij}\mathbf{U}_i^\top\mathbf{U}_j = 2\,\mathrm{Tr}(\mathbf{U}^\top\mathbf{W}\mathbf{U})$. Therefore,

$$\frac{1}{2}\sum_{i,j} W_{ij}\|\mathbf{U}_i - \mathbf{U}_j\|_2^2 = \mathrm{Tr}(\mathbf{U}^\top\mathbf{D}\mathbf{U}) - \mathrm{Tr}(\mathbf{U}^\top\mathbf{W}\mathbf{U}) = \mathrm{Tr}(\mathbf{U}^\top(\mathbf{D} - \mathbf{W})\mathbf{U}),$$

which proves Eq. 26 and PSD. □

Lemma 4 (Row-stochasticity and simplex preservation)**.** *Assume* $\mathbf{P}$ *is row-stochastic (enforced by the convention after Eq. 16). If each row of* $\tilde{\mathbf{Y}}$ *lies in the simplex* Δ^{C-1}*, then each iterate in Eq. 19 and the limit* $\mathbf{Q}$ *also have rows in* Δ^{C-1}*.*

Proof. (*Nonnegativity*) $\tilde{\mathbf{Y}} \geq 0$ entrywise and $\mathbf{P} \geq 0$ entrywise, so if $\mathbf{Q}_k \geq 0$, then $\mathbf{P}\mathbf{Q}_k \geq 0$ and $\mathbf{Q}_{k+1} = (1-\alpha)\tilde{\mathbf{Y}} + \alpha\mathbf{P}\mathbf{Q}_k \geq 0$.

(*Row sums*) Let $\mathbf{1}$ be an all-ones vector of appropriate dimension. Because $\mathbf{P}$ is row-stochastic, $\mathbf{P}\mathbf{1} = \mathbf{1}$. Assuming $\tilde{\mathbf{Y}}^{(t)}\mathbf{1} = \mathbf{1}$ (row sums are one),

$$\mathbf{Q}_{k+1}\mathbf{1} = (1-\alpha)\tilde{\mathbf{Y}}\mathbf{1} + \alpha\mathbf{P}\mathbf{Q}_k\mathbf{1} = (1-\alpha)\mathbf{1} + \alpha\mathbf{P}\mathbf{1} = \mathbf{1}.$$

Thus each row sums to one. Together with nonnegativity, rows remain in the simplex. □

4.2 Teacher Diffusion Solves a Convex Optimization Problem

Proposition 1 (Convex objective ⇒ diffusion fixed point (detailed derivation))**.** *Fix a task t (drop the superscript* (t) *for readability). Let anchors* $\tilde{\mathbf{Y}} \in \mathbb{R}^{N\times C}$ *and define* $\mathbf{W}, \mathbf{D}, \mathbf{L}$ *as in Eq. 14–15. Consider*

$$\min_{\mathbf{Q}\in\mathbb{R}^{N\times C}} J(\mathbf{Q}) := \underbrace{\frac{1-\alpha}{2}\|\mathbf{Q} - \tilde{\mathbf{Y}}\|_{\mathbf{D}}^2}_{\textit{degree-weighted fidelity}} + \underbrace{\frac{\alpha}{2}\mathrm{Tr}(\mathbf{Q}^\top\mathbf{L}\mathbf{Q})}_{\textit{motif smoothness}}, \qquad \|\mathbf{A}\|_{\mathbf{D}}^2 := \mathrm{Tr}(\mathbf{A}^\top\mathbf{D}\mathbf{A}). \tag{27}$$

Then the (unique) minimizer $\mathbf{Q}^\star$ *satisfies the diffusion equation*

$$\mathbf{Q}^\star = (1-\alpha)\tilde{\mathbf{Y}} + \alpha\mathbf{P}\mathbf{Q}^\star, \qquad \mathbf{P} = \mathbf{D}^{-1}\mathbf{W}. \tag{28}$$

Proof. **Step 1: compute $\nabla_{\mathbf{Q}} J(\mathbf{Q})$.** Write

$$J(\mathbf{Q}) = \frac{1-\alpha}{2}\mathrm{Tr}\big((\mathbf{Q}-\tilde{\mathbf{Y}})^\top \mathbf{D}(\mathbf{Q}-\tilde{\mathbf{Y}})\big) + \frac{\alpha}{2}\mathrm{Tr}(\mathbf{Q}^\top \mathbf{L}\mathbf{Q}).$$

Because $\mathbf{D}$ is symmetric,

$$\nabla_{\mathbf{Q}} \frac{1}{2}\mathrm{Tr}\big((\mathbf{Q}-\tilde{\mathbf{Y}})^\top \mathbf{D}(\mathbf{Q}-\tilde{\mathbf{Y}})\big) = \mathbf{D}(\mathbf{Q}-\tilde{\mathbf{Y}}).$$

Because $\mathbf{L}$ is symmetric (Lemma 3),

$$\nabla_{\mathbf{Q}} \frac{1}{2}\mathrm{Tr}(\mathbf{Q}^\top \mathbf{L}\mathbf{Q}) = \mathbf{L}\mathbf{Q}.$$

Thus

$$\nabla_{\mathbf{Q}} J(\mathbf{Q}) = (1-\alpha)\mathbf{D}(\mathbf{Q}-\tilde{\mathbf{Y}}) + \alpha\mathbf{L}\mathbf{Q}. \tag{29}$$

Step 2: set first-order optimality and derive a linear system. At the minimizer $\mathbf{Q}^\star$, $\nabla_{\mathbf{Q}} J(\mathbf{Q}^\star) = \mathbf{0}$:

$$(1-\alpha)\mathbf{D}(\mathbf{Q}^\star - \tilde{\mathbf{Y}}) + \alpha(\mathbf{D}-\mathbf{W})\mathbf{Q}^\star = 0.$$

Expand:

$$(1-\alpha)\mathbf{D}\mathbf{Q}^\star - (1-\alpha)\mathbf{D}\tilde{\mathbf{Y}} + \alpha\mathbf{D}\mathbf{Q}^\star - \alpha\mathbf{W}\mathbf{Q}^\star = 0.$$

Collect terms in $\mathbf{Q}^\star$:

$$\mathbf{D}\mathbf{Q}^\star - \alpha\mathbf{W}\mathbf{Q}^\star = (1-\alpha)\mathbf{D}\tilde{\mathbf{Y}}.$$

Left-multiply by $\mathbf{D}^{-1}$ (well-defined under the row-stochastic convention; isolated nodes have $P_{ii} = 1$):

$$\mathbf{Q}^\star - \alpha\mathbf{D}^{-1}\mathbf{W}\mathbf{Q}^\star = (1-\alpha)\tilde{\mathbf{Y}}.$$

Using $\mathbf{P} = \mathbf{D}^{-1}\mathbf{W}$ yields

$$\mathbf{Q}^\star = (1-\alpha)\tilde{\mathbf{Y}} + \alpha\mathbf{P}\mathbf{Q}^\star,$$

which is Eq. 28.

Step 3: uniqueness. $J(\mathbf{Q})$ is a quadratic. The smoothness term is convex because $\mathbf{L} \succeq 0$ (Lemma 3). The fidelity term is strictly convex on rows with $D_{ii} > 0$. With the convention $P_{ii} = 1$ for isolated nodes, those rows satisfy $\mathbf{Q}_i = \tilde{\mathbf{Y}}_i$ uniquely under Eq. 18. Hence the solution is unique. □

4.3 Closed Form and Neumann Series Expansion

Proposition 2 (Closed form and Neumann series)**.** *Assume $\mathbf{P}$ is row-stochastic and $\alpha \in (0,1)$. Then Eq. 18 has the unique solution*

$$\mathbf{Q}^\star = (1-\alpha)(\mathbf{I}-\alpha\mathbf{P})^{-1}\tilde{\mathbf{Y}} = (1-\alpha)\sum_{k=0}^{\infty}(\alpha\mathbf{P})^k\tilde{\mathbf{Y}}. \tag{30}$$

Proof. Rearrange Eq. 18:

$$(\mathbf{I}-\alpha\mathbf{P})\mathbf{Q}^\star = (1-\alpha)\tilde{\mathbf{Y}}.$$

Because $\mathbf{P}$ is row-stochastic and nonnegative, its induced ℓ_∞ operator norm satisfies $\|\mathbf{P}\|_\infty = 1$ (maximum absolute row sum). Thus $\|\alpha\mathbf{P}\|_\infty = \alpha < 1$. Therefore the Neumann series $\sum_{k=0}^{\infty}(\alpha\mathbf{P})^k$ converges and equals $(\mathbf{I}-\alpha\mathbf{P})^{-1}$, yielding Eq. 30. Invertibility implies uniqueness. □

4.4 Convergence of Power Iteration (Explicit Error Bound)

Proposition 3 (Contraction and linear rate)**.** *Define $\mathcal{T}(\mathbf{Q}) = (1-\alpha)\tilde{\mathbf{Y}} + \alpha\mathbf{P}\mathbf{Q}$. If $\mathbf{P}$ is row-stochastic and $\alpha \in (0,1)$, then for any $\mathbf{Q}, \mathbf{Q}'$,*

$$\|\mathcal{T}(\mathbf{Q}) - \mathcal{T}(\mathbf{Q}')\|_\infty \le \alpha\|\mathbf{Q}-\mathbf{Q}'\|_\infty. \tag{31}$$

Let $\mathbf{Q}^\star$ be the fixed point. Then the iterates in Eq. 19 satisfy

$$\|\mathbf{Q}_k - \mathbf{Q}^\star\|_\infty \le \alpha^k\|\mathbf{Q}_0 - \mathbf{Q}^\star\|_\infty. \tag{32}$$

Proof. We have

$$\mathcal{T}(\mathbf{Q}) - \mathcal{T}(\mathbf{Q}') = \alpha \mathbf{P}(\mathbf{Q} - \mathbf{Q}').$$

Because $\mathbf{P}$ is row-stochastic and nonnegative, for any matrix $\mathbf{A}$, each entry of $\mathbf{PA}$ is a convex combination of entries of $\mathbf{A}$ in the corresponding column, hence $\|\mathbf{PA}\|_\infty \leq \|\mathbf{A}\|_\infty$. Therefore

$$\|\mathcal{T}(\mathbf{Q}) - \mathcal{T}(\mathbf{Q}')\|_\infty = \alpha \|\mathbf{P}(\mathbf{Q} - \mathbf{Q}')\|_\infty \leq \alpha \|\mathbf{Q} - \mathbf{Q}'\|_\infty,$$

which proves Eq. 31. Applying the standard fixed-point iteration bound for contractions yields Eq. 32. □

4.5 Distillation: KL vs. Soft Cross-Entropy (Fully Expanded)

Proposition 4 (Categorical KL equals soft CE plus a constant)**.** *Let* $\mathbf{q}, \mathbf{p} \in \Delta^{C-1}$ *and define categorical distributions* $Q = \mathrm{Cat}(\mathbf{q})$*,* $P = \mathrm{Cat}(\mathbf{p})$*. Then*

$$\mathrm{KL}(Q\|P) = \sum_{c=1}^{C} q_c \log \frac{q_c}{p_c} = -\sum_{c=1}^{C} q_c \log p_c + \sum_{c=1}^{C} q_c \log q_c = \mathrm{CE}(\mathbf{q}, \mathbf{p}) - H(\mathbf{q}), \tag{33}$$

where $H(\mathbf{q}) = -\sum_c q_c \log q_c$ *is independent of* $\mathbf{p}$ *(hence independent of* θ*). Therefore minimizing* $\mathrm{KL}(Q\|P_\theta)$ *over* θ *is equivalent to minimizing* $\mathrm{CE}(\mathbf{q}, \mathbf{p}_\theta)$*.*

Proof. The equality follows by direct expansion of $\mathrm{KL}(Q\|P)$:

$$\sum_c q_c \log \frac{q_c}{p_c} = \sum_c q_c \log q_c - \sum_c q_c \log p_c.$$

Recognize $-\sum_c q_c \log p_c = \mathrm{CE}(\mathbf{q}, \mathbf{p})$ and $\sum_c q_c \log q_c = -H(\mathbf{q})$. Since $H(\mathbf{q})$ does not depend on θ, it can be dropped during optimization. □

4.6 Inductive Inference (No Transductive Leakage)

Proposition 5 (Inductive inference)**.** *Let* $\theta^\star$ *be the output of training. For any test molecule* G_{test}*, MotifDiff-EM outputs* $\mathbf{p}_{\text{test},t} = p_{\theta^\star}(\cdot \mid G_{\text{test}}, t)$*. No diffusion graph containing* G_{test} *is constructed and no inter-molecule propagation is performed at test time.*

Proof. Algorithm 1 returns only the student parameters $\theta^\star$. Teacher diffusion is used solely during training to compute $\mathbf{Q}^{(t)}$ on molecules in the training pool $\{1, \ldots, N\}$. At inference, the predictor is the local mapping $G \mapsto p_{\theta^\star}(\cdot \mid G, t)$, which depends only on G and $\theta^\star$. Hence inference is inductive and cannot exploit relationships among test molecules. □

4.7 Interpretability from Motif Contributions

Eq. 22 provides neighbor-molecule contributions $\alpha P_{ij} Q_{jc}$. Eq. 25 provides motif-level contributions s_{imc}, which can be used to rank motifs for explanation. Reliability weights r_m further improve robustness by suppressing motifs with high labeled inconsistency, mitigating error amplification under activity-cliff violations, while preserving convexity and convergence (Lemmas 2–3).

5 EXPERIMENTS

6 CONCLUSION

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A Shared Reliability of Multi-task with Missing Labels

For task t with C_t classes, define $H_m^{(t)}$ using only molecules with observed labels for task t (i.e., $i \in \mathcal{L}_t$). We share a single reliability by averaging entropies:

$$\bar{H}_m = \frac{\sum_{t=1}^{T} \mathbb{I}[n_m^{(t)} > 0] \cdot H_m^{(t)}}{\sum_{t=1}^{T} \mathbb{I}[n_m^{(t)} > 0] + \epsilon}, \qquad r_m = \exp(-\gamma \bar{H}_m), \tag{34}$$

where $n_m^{(t)}$ is the number of labeled occurrences of motif m for task t. (Optionally, a task-specific $r_m^{(t)}$ can be used at higher compute cost; we keep the shared version for minimal change.)

B Regression Instantiation

We instantiate MotifDiff-EM for regression tasks. The motif graph $(\mathbf{B}, \mathbf{W}, \mathbf{P})$ and diffusion teacher are unchanged; only anchors, matching loss, and gating are adapted.

B.1 Anchors and Diffusion

For regression, anchors are scalars:

$$\tilde{y}_i = \begin{cases} y_i, & i \in \mathcal{D}_L, \\ \mu_\theta(G_i), & i \in \mathcal{D}_U, \end{cases}$$

where $\mu_\theta(G)$ is the student-predicted mean. The same diffusion fixed point

$$\mathbf{q} = (1-\alpha)\tilde{\mathbf{y}} + \alpha \mathbf{P}\mathbf{q}$$

yields teacher targets $q_i \in \mathbb{R}$.

B.2 Gaussian Distillation and MSE Equivalence

Assume a homoscedastic Gaussian likelihood $p_\theta(y \mid G) = \mathcal{N}(y; \mu_\theta(G), \sigma^2)$. Distillation minimizes Gaussian NLL on unlabeled pairs:

$$\min_\theta \; \mathcal{L}_{\text{sup}}^{\text{reg}}(\theta) + \lambda \sum_{i \in \mathcal{D}_U : g_i = 1} \left[-\log \mathcal{N}(q_i; \mu_\theta(G_i), \sigma^2) \right], \tag{35}$$

where $\mathcal{L}_{\text{sup}}^{\text{reg}}$ is the supervised regression loss on $\mathcal{D}_L$ (e.g., Gaussian NLL or MSE), and g_i is a regression gating rule. Because regression lacks max-probability confidence, use either: (i) heteroscedastic variance $\sigma_\theta^2(G)$ and gate by $\sigma_\theta^2(G) \le \kappa$, or (ii) consistency gating from two stochastic passes and gate by $|\mu_\theta^{(1)}(G) - \mu_\theta^{(2)}(G)| \le \kappa$.

Proposition 6 (Gaussian NLL $\Leftrightarrow$ MSE (homoscedastic))**.** *If σ^2 is constant, minimizing $-\log \mathcal{N}(q; \mu, \sigma^2)$ w.r.t. μ is equivalent (up to constants and scaling) to minimizing $(q-\mu)^2$.*

Proof. $-\log \mathcal{N}(q; \mu, \sigma^2) = \frac{(q-\mu)^2}{2\sigma^2} + \frac{1}{2}\log \sigma^2 + \text{const}$. Dropping constants and scaling yields MSE. □